

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EC) Examination

November 2013

EC405 Digital System Design

Date: 23.11.2013, Saturday Time: 10.00a.m. To 1.00p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION-I

Q-1 Attempt the following.

- 1 An N-input AND with a constant 1 value on one input is logically equivalent to the _____ input AND gate. 01
- 2 If the dividend register initially contained 001101011 and divisor is 0101, can division take place? Explain. 02
- 3 Using KMAP, explain why $F = a'b + ac$ has a 1 hazard for the input change $abc = 011$ to 111 but not for 011 to 010 . 02
- 4 A Mealy sequential network with four output variables is realized using 22V10. What is the maximum number of input variables it can have? Can any Mealy network with these numbers of inputs and outputs be realized with a 22V10? Justify. 02

- Q-2(a) A sequential network has one input (x) and one output (z). The output $z=1$ if the total number of 1's received is odd and at least two consecutive 0's have been received. Find Moore state graph. 04
- (b) In order to allow for a larger number of bits, the control circuit for binary multiplier is to be redesigned so that it uses a separate counter and add shift control. Draw the state graph for the add-shift control. 04
- (c) Find a minimum row PLA table to implement a following sets of function: 06
- $F_1 = \sum m(2,3,5,7,8,9,10,11,13,15)$
 $F_2 = \sum m(2,3,5,6,7,10,11,14,15)$
 $F_3 = \sum m(6,7,8,9,13,14,15)$

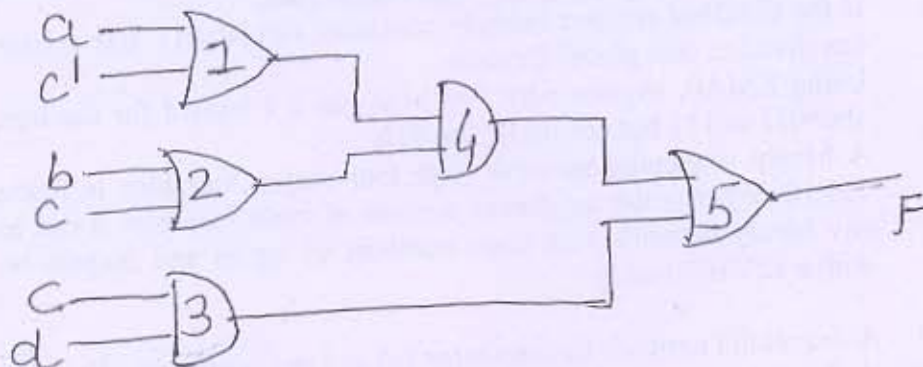
OR

- Q-2 (a) A sequential network has one input (x) and one output (z). The network examines groups of four consecutive inputs and produces an output $z=1$ if the input sequence 0101 or 1001 occurs. The network resets after every four inputs. Find the Mealy state graph. 04
- (b) What is Fault Equivalence? Find out fault equivalence class for below mentioned gates: NAND, NOR 04
- (c) Using the method of map entered variables, use 4 variable maps to find a minimum sum of product expression. Also verify it with 5 variable kmap 06
- $F(A,B,C,D,E) = \sum m(0,4,6,13,14) + \sum d(2,9) + E(1,12).$

- Q-3(a) Draw and explain the SM chart for UART transmitter. 06
- (b) A store has one express register and three regular ones. It is the store policy that the express register be open only when two or more of the other registers are busy. Assume that the Boolean variables A,B & C reflect the status of each of the regular registers (1 busy, 0 idle). Design the logic circuit, with A,B & C as inputs & F as output, to automatically notify the manager (by selecting F=1) to open the express register. 06
- For the circuit, find the set of all tests that detect the fault A s-a-0.
- (c) What is the main difference between a PAL- and a GAL-based SPLD? 02

OR

- Q-3 (a) Design binary divider to divide 6 bit dividend by a 3 bit divisor to find a 3 bit quotient. The right 3 bit of the dividend register should be used to store the quotient. Draw a block diagram for the divider along with state graph. 06
- (b) Find out static 1 and static 0 hazards in the network given below if any. 06



- (c) Write a VHDL description of JK flip-flop using behavioral modeling style. 02

SECTION II

- Q-4 Do as directed.
- 1 Which of the following(s) is not applicable when treating hazards? 02
- (1) $X+X'=1$
 (2) $XX=X$
 (3) $XX'=0$
 (4) None
- 2 Define the following terms: 02
- Bridging faults, Signature analysis
- 3 A wrong output signal produced by a defective system is called an/a 01
- _____
- 4 Which are the different types of pattern generators for BIST architecture? 02
- Q-5(a) Draw and explain the complete flow for digital design. Highlight the stages for which it will be different from programmable logic flow in flow chart. 06
- (b) Given the polynomial $P(x) = x^4 + x^1 + 1$. 05
- (i) Design a LSFR with characteristic polynomial $P(x)$.
- (ii) Initial state of the LFSR is "1000" then determine the complete

sequence.

- (iii) Add logic so that "0000" is included in the state sequence.

- (c) Construct an SM chart that has 3 input variables (D, E, F), 4 output variables (P, Q, R, S), and 2 exit paths. For this block, output P is always 1, and Q is 1 if D=1. If D and F are 1 or if D and E are 0, R=1 and exit path 2 is taken. If (D=0 and E=1) or (D=1 and F=0), S=1 and exit path 1 is taken. 03

OR

- Q-5(a) What is BILBO? Describe BILBO architecture and its operation? 06

- (b) Design sequential traffic light controller for the intersection of a street "A" and "B" street with state graph. 05

- (c) Alyssa P. Hacker has a snail that crawls down a paper tape with 1's and 0's on it. The snail smiles whenever the last three digits it has crawled over are 101. Design Moore FSMs of the snail's brain and draw SM chart for the same. 03

- Q-6(a) Compare the different programming technologies used in PLD. 04

- (b) Draw the block diagram of keypad scanner. Which are the main modules of keypad scanner? Explain the working of keyscan using state graph. 07

- (c) A clocked mealy sequential network with one input (X) and one output (Z) is to be designed. The output is to be 0, unless the input is 0 following sequence of exactly two inputs followed by a 1 input. 03

Specify the output sequence for each of the following input sequences.

[1] x=0010

[2] x=.....10010

[3] x=.....00010[...means input sequence not ending in 00]

OR

- (a) Draw a block diagram of a 4 bit serial adder with Accumulator. Explain operation of it and also describe control state graph and design its control circuit. 06

- (b) Explain the SM chart for Dice game and implement it using PLA and flip-flop. 08